

# A Complete Course in Physics ( Graphs ) - First Edition

Rajat Kalia

July 30, 2012



# Contents

|          |                                                                                                                 |          |
|----------|-----------------------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Concepts of Graphs</b>                                                                                       | <b>7</b> |
| 1.1      | The Equations of motion and the origin of Graph Handling . . . . .                                              | 7        |
| 1.1.1    | The First Equation . . . . .                                                                                    | 7        |
| 1.1.2    | The Second Equation . . . . .                                                                                   | 8        |
| 1.1.3    | The Acceleration-Position Graph Variate . . . . .                                                               | 11       |
| 1.2      | Other Types of Graphs . . . . .                                                                                 | 12       |
| 1.2.1    | Sign of Acceleration from Position-Time Graph . . . . .                                                         | 12       |
| 1.2.2    | The Average-Velocity / Instantaneous Velocity , Equal Case . . . . .                                            | 13       |
| 1.2.3    | The Velocity-Displacement Case . . . . .                                                                        | 13       |
| 1.2.4    | Motion Under Free Fall due to Gravity . . . . .                                                                 | 15       |
| 1.2.5    | Projectile Motion . . . . .                                                                                     | 16       |
| 1.2.6    | Miscellaneous . . . . .                                                                                         | 17       |
| 1.3      | Question Types . . . . .                                                                                        | 21       |
| 1.3.1    | Passage Type . . . . .                                                                                          | 21       |
| 1.3.2    | Matching . . . . .                                                                                              | 23       |
| 1.4      | Previous Year Problems IIT . . . . .                                                                            | 24       |
|          | Infrastructure                                                                                                  |          |
|          | #764, 41-A, Chandigarh {TC(official)} - 160036                                                                  |          |
|          | Shamlat (Ramgarh Sardaran), Ludhiana, Punjab                                                                    |          |
|          | SCO 24(TF), 41-D, Chandigarh - 160036                                                                           |          |
|          | <a href="https://www.alphaclasses.com/">https://www.alphaclasses.com/</a>                                       |          |
|          | Educational                                                                                                     |          |
|          | <a href="https://github.com/rajatkalia1982">https://github.com/rajatkalia1982</a>                               |          |
|          | <a href="https://sourceforge.net/u/rajatkalia/profile">https://sourceforge.net/u/rajatkalia/profile</a>         |          |
|          | <a href="https://www.youtube.com/@rajatkaliaeducation">https://www.youtube.com/@rajatkaliaeducation</a>         |          |
|          | Store                                                                                                           |          |
|          | <a href="https://www.lulu.com/spotlight/rajatkalia11">https://www.lulu.com/spotlight/rajatkalia11</a>           |          |
|          | <a href="https://amazon.com/author/rajatkalia">https://amazon.com/author/rajatkalia</a>                         |          |
|          | Sports                                                                                                          |          |
|          | <a href="https://www.youtube.com/@indiagamersorganization">https://www.youtube.com/@indiagamersorganization</a> |          |
|          | <a href="https://www.facebook.com/groups/igohn">https://www.facebook.com/groups/igohn</a>                       |          |
|          | <a href="https://india.voobly.com/">https://india.voobly.com/</a>                                               |          |
|          | Social                                                                                                          |          |
|          | <a href="https://www.goodreads.com/sardarbhagatsingh">https://www.goodreads.com/sardarbhagatsingh</a>           |          |
|          | <a href="https://x.com/RajatKalia1">https://x.com/RajatKalia1</a>                                               |          |
|          | <a href="https://www.linkedin.com/in/rajatkalia007/">https://www.linkedin.com/in/rajatkalia007/</a>             |          |
|          | <a href="https://www.facebook.com/rajatkalia007">https://www.facebook.com/rajatkalia007</a>                     |          |
|          | <a href="https://www.instagram.com/rajatkalia007/">https://www.instagram.com/rajatkalia007/</a>                 |          |
|          | <a href="https://www.twitch.tv/rajatkalia">https://www.twitch.tv/rajatkalia</a>                                 |          |
|          | <a href="https://www.facebook.com/rajatkaliaeducation">https://www.facebook.com/rajatkaliaeducation</a>         |          |
|          | <a href="https://www.researchgate.net/profile/Rajat-Kalia">https://www.researchgate.net/profile/Rajat-Kalia</a> |          |
|          | <a href="https://www.w3profile.com/rajatkalia/">https://www.w3profile.com/rajatkalia/</a>                       |          |
|          | <a href="https://www.reddit.com/user/rajatkalia/">https://www.reddit.com/user/rajatkalia/</a>                   |          |
|          | Blogs                                                                                                           |          |
|          | <a href="https://rk-e.blogspot.in/">https://rk-e.blogspot.in/</a> [Physics]                                     |          |
|          | <a href="https://chem-rke.blogspot.com/">https://chem-rke.blogspot.com/</a> [Chemistry]                         |          |
|          | <a href="https://biosciences-rke.blogspot.com/">https://biosciences-rke.blogspot.com/</a> [ Bio-Sciences ]      |          |
|          | <a href="https://rk-edu.blogspot.com/">https://rk-edu.blogspot.com/</a> [ Mathematics ]                         |          |
|          | <a href="https://medlife-rke.blogspot.com/">https://medlife-rke.blogspot.com/</a> [Medical Sciences ]           |          |

<https://computers-mke.blogspot.com/> [ Computers Engineering ]  
<https://ee-mke.blogspot.com/> [Electrical Engineering ]  
<https://me-mke.blogspot.com/> [ Mechanical Engineering ]  
<https://ce-mke.blogspot.com/> [ Civil Engineering ]  
<https://mke-whj.blogspot.com/> [ White Hat Junior ]  
<https://ba-xpn.blogspot.com/> [ Bachelor of Arts ]  
<https://constitution1982.blogspot.com/> [ Constitution ]  
<https://debk-mke.blogspot.com/> [ Accounting ]  
<https://comm-pke.blogspot.com/> [ Commerce ]  
<https://brigade-13.blogspot.com/> [ Air Force, India ]  
<https://battalion-7.blogspot.com/> [ Armed Forces ]  
<https://design-mke.blogspot.com/> [ Design ]  
<https://tmm18.blogspot.com/> [ Meditation ]  
Freelancing  
<https://www.guru.com/freelancers/rajat-kalia>  
<https://www.codecademy.com/profiles/rajatkalia>  
<https://leetcode.com/u/rajatkalia1/>  
<https://www.scribd.com/user/308680256/Rajat-Kalia>

# Preface and Acknowledgements

This book is a research evaluate stuff on Kinematics covering graphs. The material has got acclaim locally and now looking for it internationally.

I would like to thank my family, well wishers for their support and consistent encouragement and most of all God for it's grace and Guidance.



# Chapter 1

## Concepts of Graphs

### 1.1 The Equations of motion and the origin of Graph Handling

#### 1.1.1 The First Equation

The Equation  $v = \frac{dx}{dt}$  in linear motion implies

- i) The **Slope** of **Position-Time Graph** is **Instantaneous Velocity**.
- ii) The **Area** under the **Velocity-Time Graph** is **Change in Position**.  
{ The second one requires the manipulation ,  $dx = vdt$  i.e.  $\int dx = \int vdt$  }

The equations can be further manipulated to obtain the Speed Time Graph , where

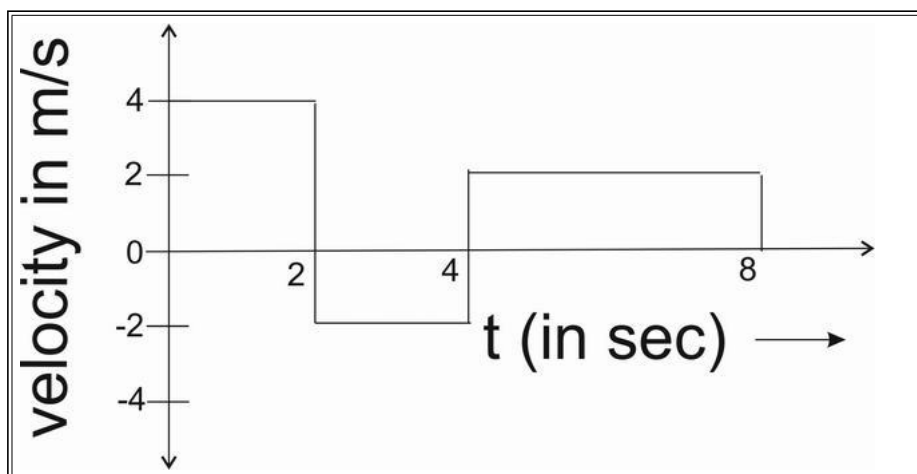
$$\text{speed} = \text{rate of change of distance wrt time}$$

Few of the following examples illustrate this concept:

**Example:** On a displacement-time graph, two straight lines make angles of  $30^\circ$  and  $60^\circ$  with the time-axis.  
The ratio of the velocities represented by them is

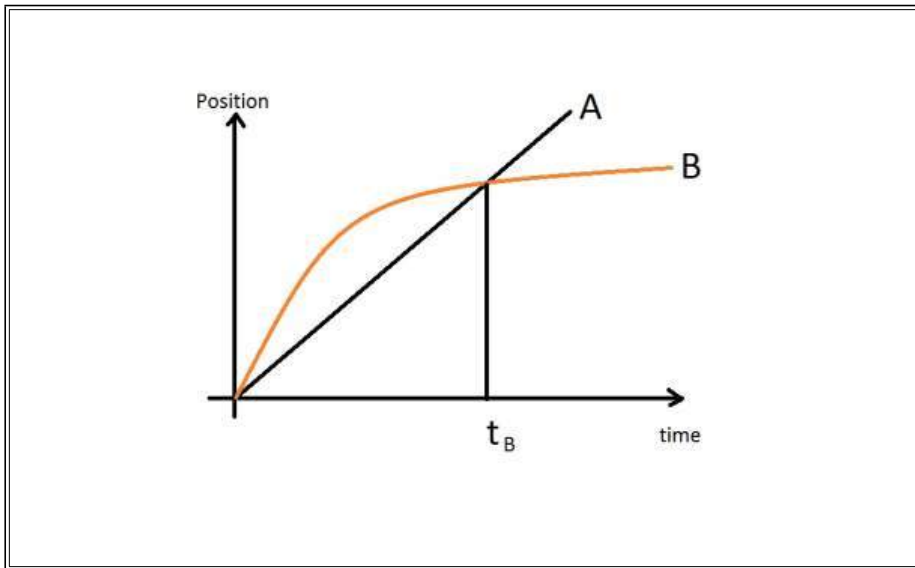
- a)  $1 : \sqrt{3}$
- b)  $1:3$
- c)  $\sqrt{3} : 1$
- d)  $3:1$

**Example:** A body is moving in a straight line as shown in velocity-time graph. The displacement and distance travelled by body in 8 second are respectively:



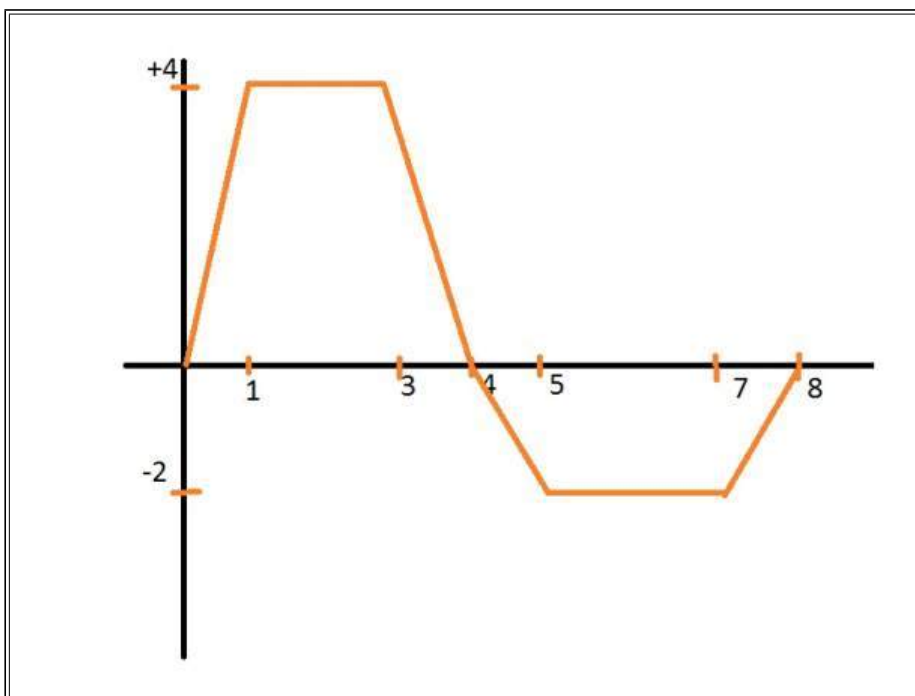
- a) 12 m, 20 m
- b) 20 m, 12 m
- c) 12 m, 12 m
- d) 20 m, 20 m

**Example:** The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?



- a) At time  $t_B$  both trains have the same velocity.
- b) Both trains have the same velocity at some time after  $t_B$ .
- c) Both trains have the same velocity at some time before  $t_B$ .
- d) Somewhere on the graph, both trains have the same acceleration.

**Example:** The velocity-time graph of a particle in linear motion is as shown. Both  $v$  and  $t$  are in SI units. The displacement of the particle is



- a) 6 m
- b) 8 m
- c) 16 m
- d) 18 m

### 1.1.2 The Second Equation

Proceeding similar to above, the equation  $a = \frac{dv}{dt}$  implies



i) The **Slope** of **Velocity-Time Graph** is **Instantaneous Acceleration**.

ii) The **Area** under **Acceleration-Time Graph** is **Change in Velocity**.

{ The second one requires the manipulation ,  $dv = a dt$  i.e.  $\int dv = \int a dt$  }

A few of the following examples illustrate it.

**Example:** A car starts from rest acquires a velocity  $v$  with uniform acceleration  $2ms^{-2}$  then it comes to stop with uniform retardation  $4ms^{-2}$  . If the total time for which it remains in motion is 3 sec, the total distance travelled is:

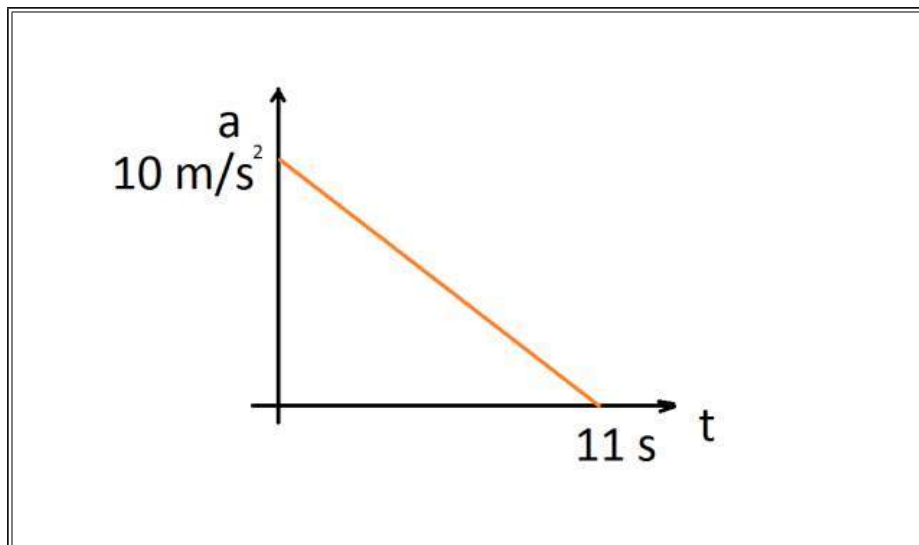
a) 2 m

b) 3 m

c) 4 m

d) 6 m

**Example:** A particle starts from rest. Its acceleration ( $a$ ) vs time ( $t$ ) is as shown in the Figure. The maximum speed of the particle will be



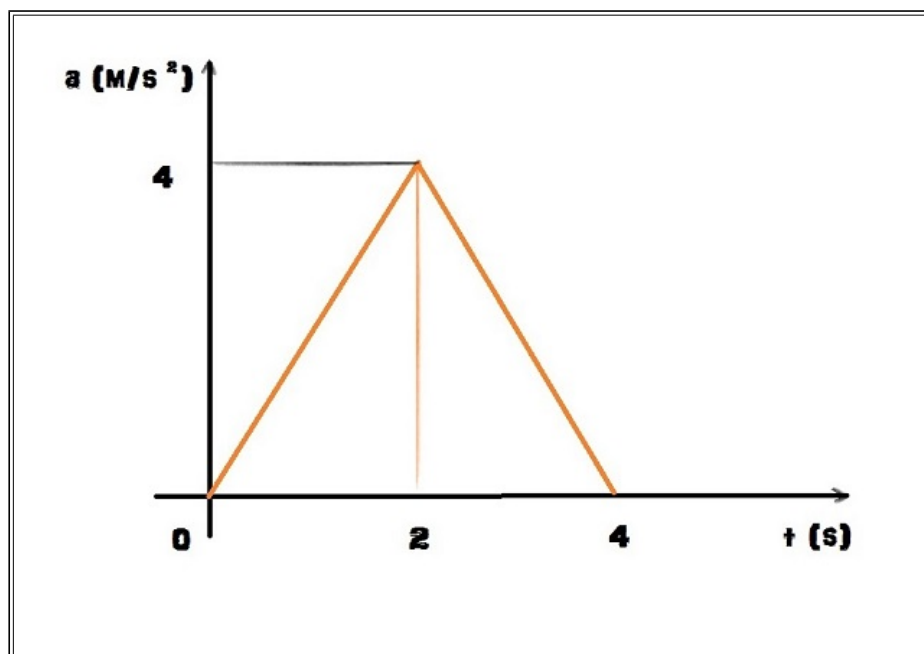
a) 110 m/s

b) 55 m/s

c) 550 m/s

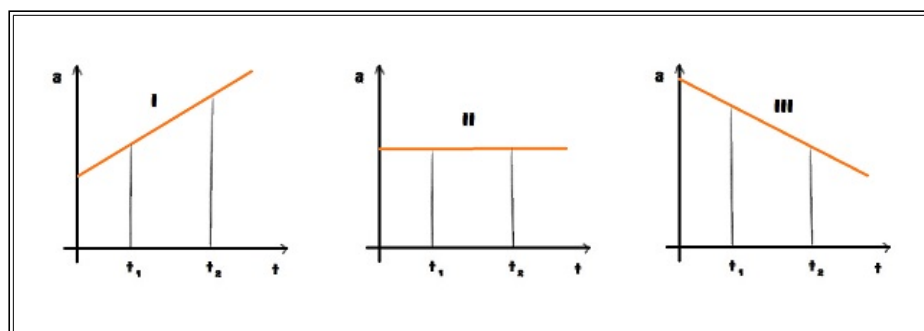
d) 660 m/s

**Example:** Acceleration-time graph of a particle moving in a straight line is shown in Figure. The velocity of particle at time  $t = 0$  is 2 m/s. Velocity at the end of fourth second is



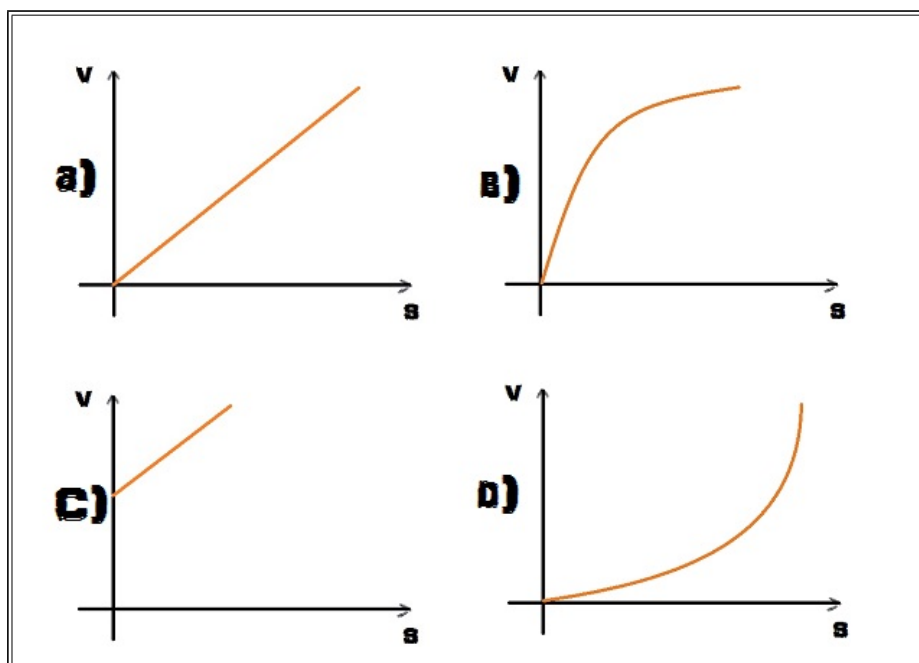
- a) 8 m/s
- b) 10 m/s
- c) 12 m/s
- d) 14 m/s

**Example:** Each of the three graphs represents acceleration vs time for an object that already has a positive velocity at time  $t_1$ . Which graph/graphs show an object whose speed is increasing for the entire time interval between  $t_1$  and  $t_2$ ?



- a) Graph I only
- b) Graphs I and II
- c) Graphs I and III
- d) Graphs I, II and III

**Example:** A body starts from rest and moves along a straight line with constant acceleration. The variation of speed  $v$  with distance  $s$  is given by the graph



### 1.1.3 The Acceleration-Position Graph Variate

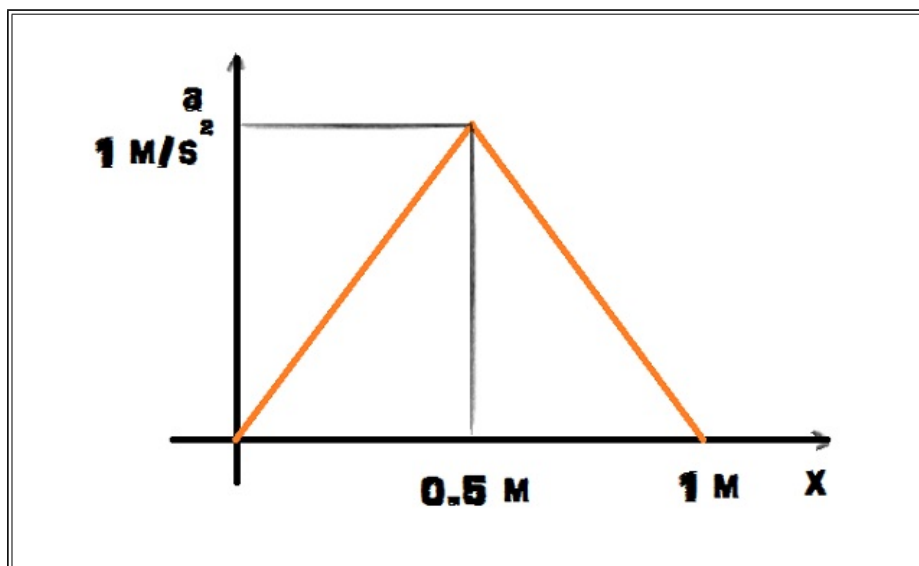
This kind of graph requires the manipulation of the Equation  $a = \frac{dv}{dt}$  as follows

$$a = \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$\Rightarrow a = \frac{dv}{dx} \cdot v$$

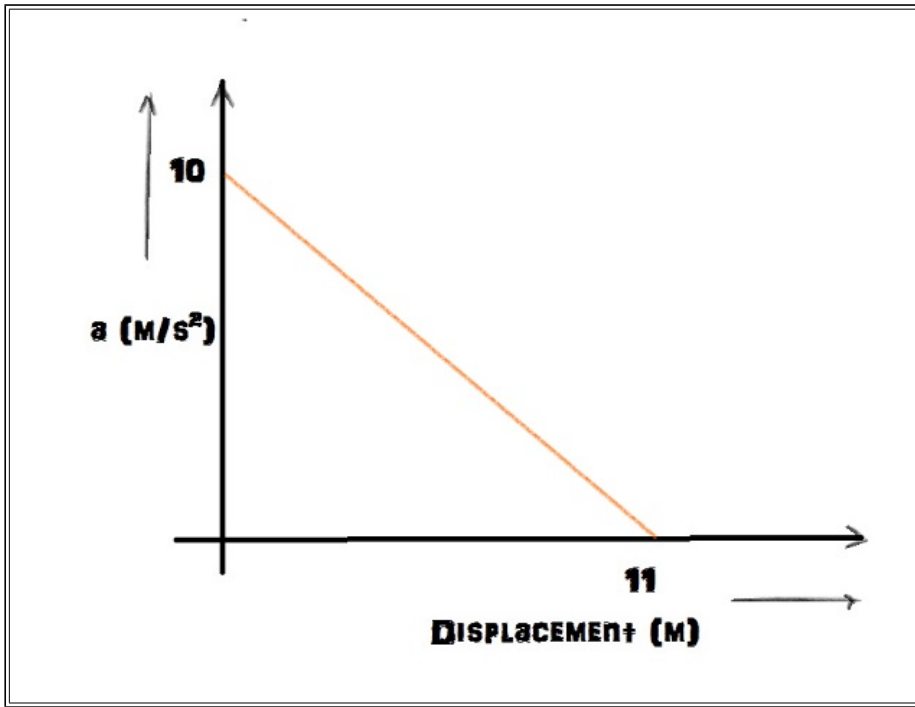
$\Rightarrow adx = vdv$  and integration can be performed to further solve it.

**Example:** A body, initially at rest, starts moving along x-axis in such a way that its acceleration vs displacement plot is as shown in the Figure. The maximum velocity of the particle is



- a) 1 m/s
- b) 6 m/s
- c) 2 m/s
- d) None of these

**Example :** A particle initially at rest, it is subjected to a non-uniform acceleration  $a$ , as shown in the gure. The maximum speed attained by the particle is



- a) 605 m/s
- b) 110 m/s
- c) 55 m/s
- d) 110 m/s

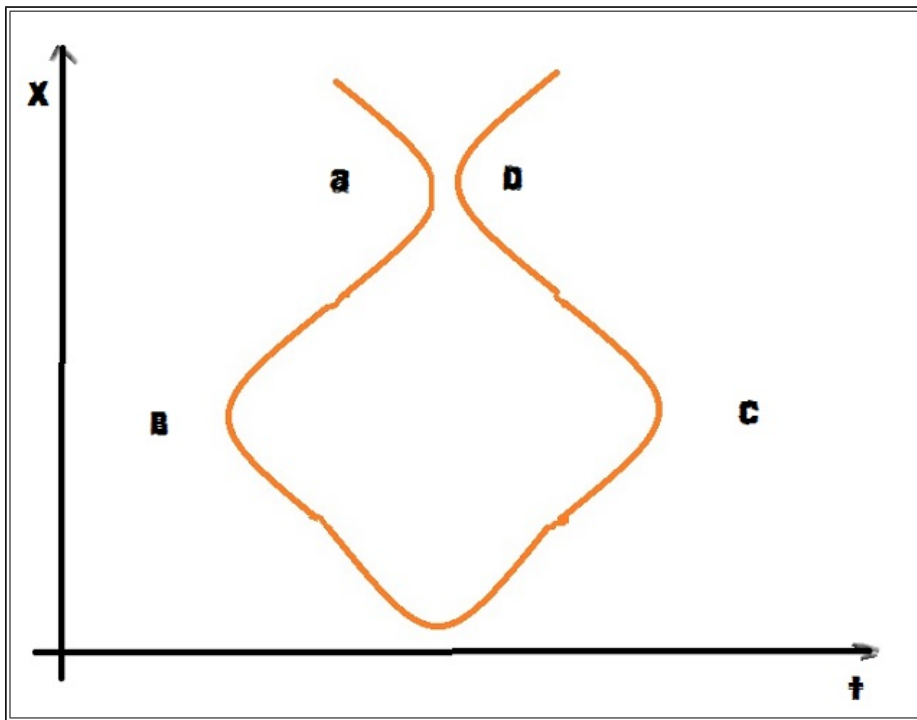
## 1.2 Other Types of Graphs

### 1.2.1 Sign of Acceleration from Position-Time Graph

The sign of Acceleration can be determined from the Position-Time Graph. The methodology involves looking at the Concavity of the Graph

- i) If the graph is Concave-Up, the Acceleration is Positive.
- ii) If the graph is Concave-Down, the Acceleration is Negative.
- iii) If the graph is a straight line, the Acceleration is ZERO. { Irrespective of any other factor , such as the slope or direction of line }

**Example:** The graph given below is a plot of distance vs time. For which labelled region is the “Velocity Positive and the Acceleration Negative”

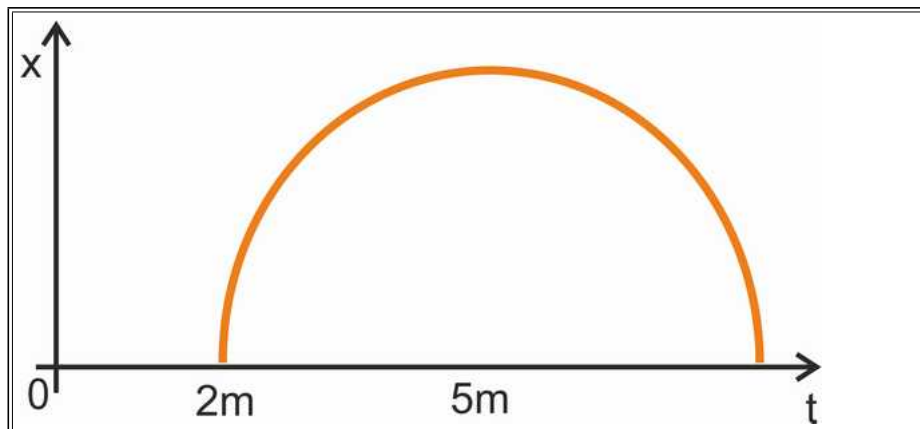


- a) a
- b) b
- c) c
- d) d

### 1.2.2 The Average-Velocity / Instantaneous Velocity , Equal Case

We know , that ( in a x-t graph) the slope of the Secant is the Average Velocity , whereas the slope of Tangent is the Instantaneous Velocity. The point where these two lines coincide, is the point where Average Velocity is equal to Instantaneous Velocity.

**Example:** Position-time graph is shown which is a semicircle from  $t = 2$  to  $t = 8$  s. Find time  $t$  at which the instantaneous velocity is equal to average velocity over first  $t$  seconds,



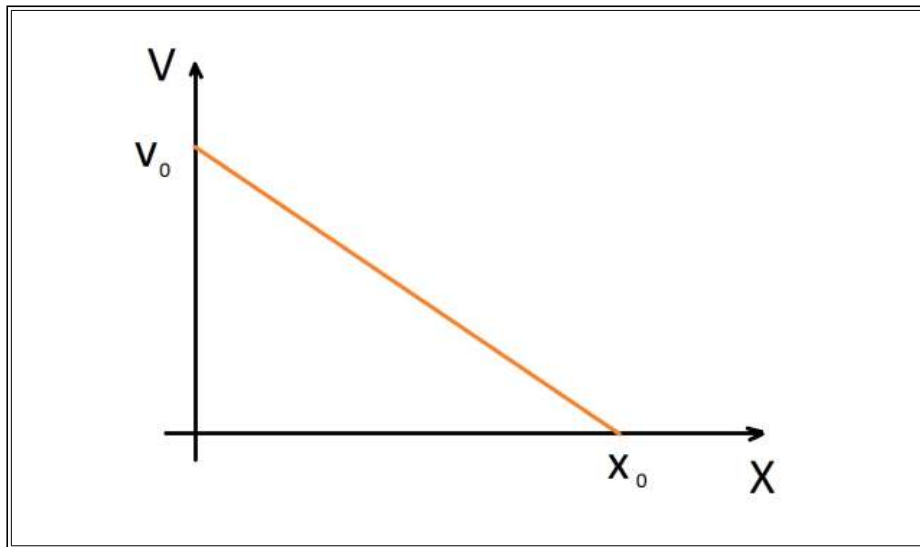
- a) 4.8 s
- b) 3.2 s
- c) 2.4 s
- d) 5 s

### 1.2.3 The Velocity-Displacement Case

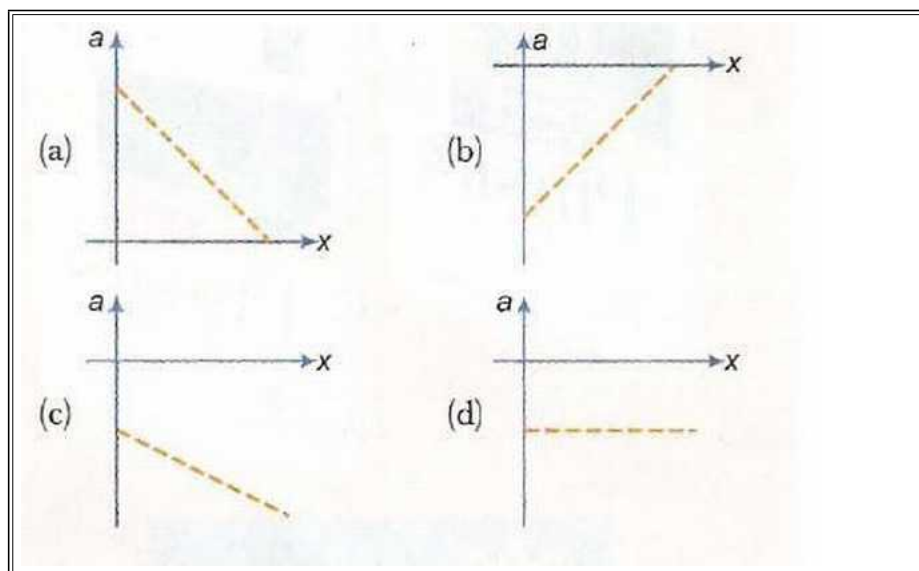
This can be handled in a similar way as Acceleration-Displacement case by integrating the respective equation. Here the problem is of  $v = f(x)$  type, which can be integrated by writing  $\frac{dx}{dt} = f(x)$

i.e.  $dx = f(x)dt$

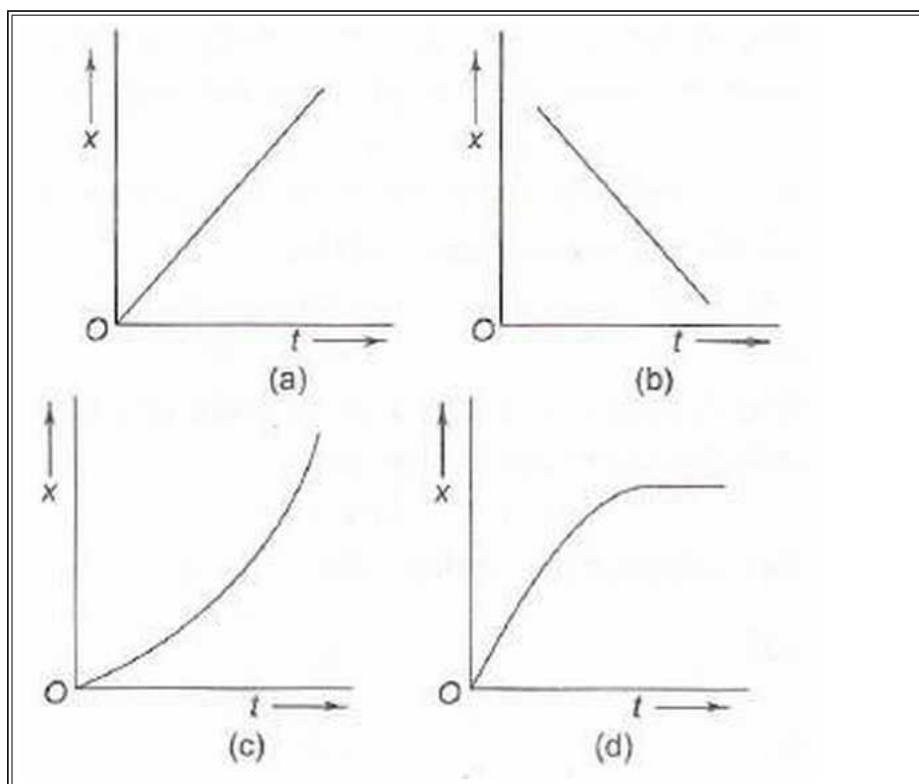
**Example:** The velocity-displacement graph of a particle moving along a straight line is shown here.



The most suitable acceleration-displacement graph will be



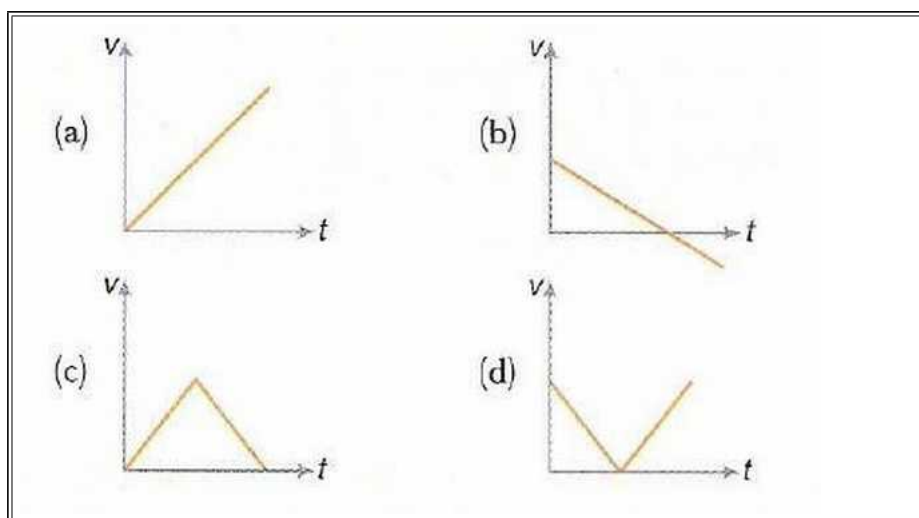
**Example.** The velocity ( $v$ ) of a body moving along the positive  $x$ -direction varies with displacement ( $x$ ) from the origin as  $p v = k x$ , where  $k$  is a constant. Which of the graphs shown in Fig. correctly represents the displacement-time ( $x - t$ ) graph of the motion?



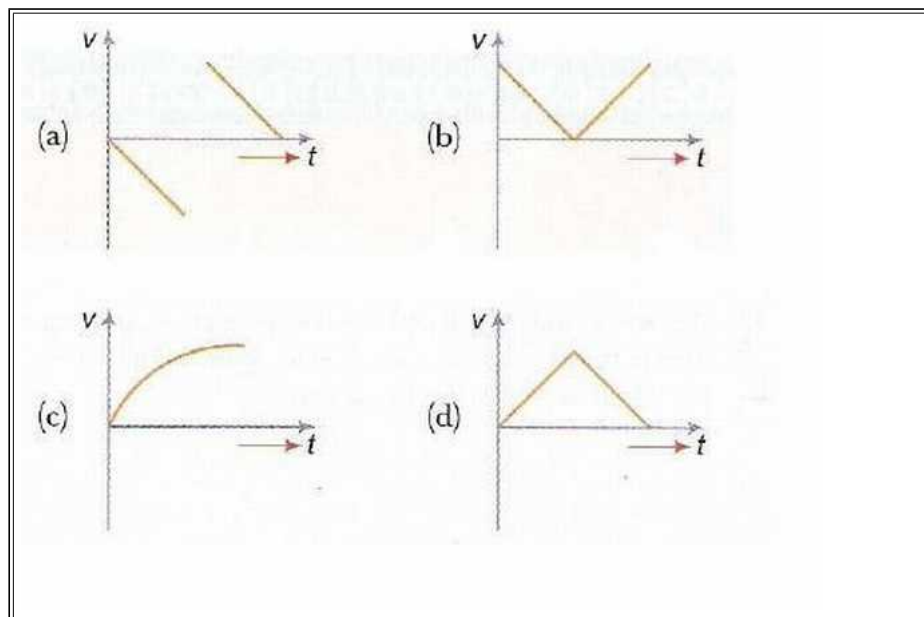
### 1.2.4 Motion Under Free Fall due to Gravity

In such examples, the governing equations rule and the coordinate system needs to be properly chosen.

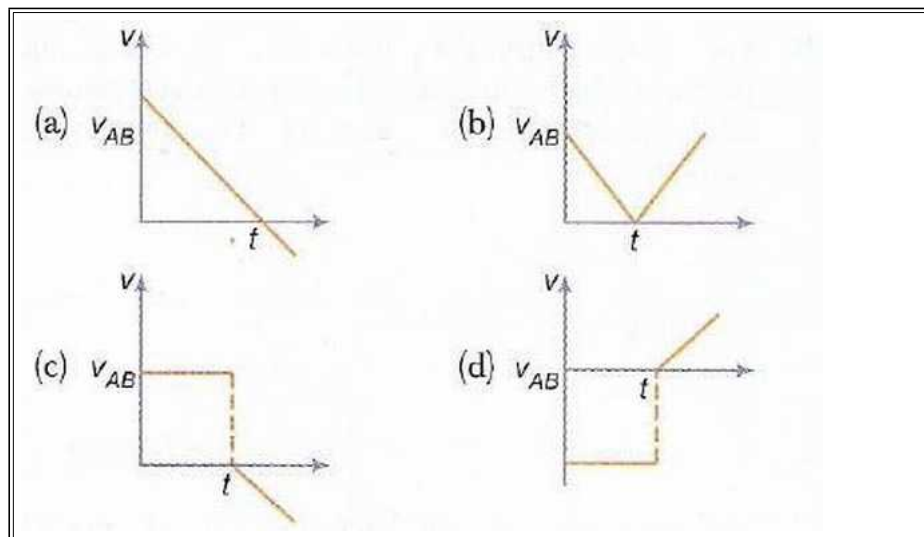
**Example:** Which of the following graphs correctly represents velocity-time relationship for a particle released from rest to fall freely under gravity?



**Example:** The velocity-time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?



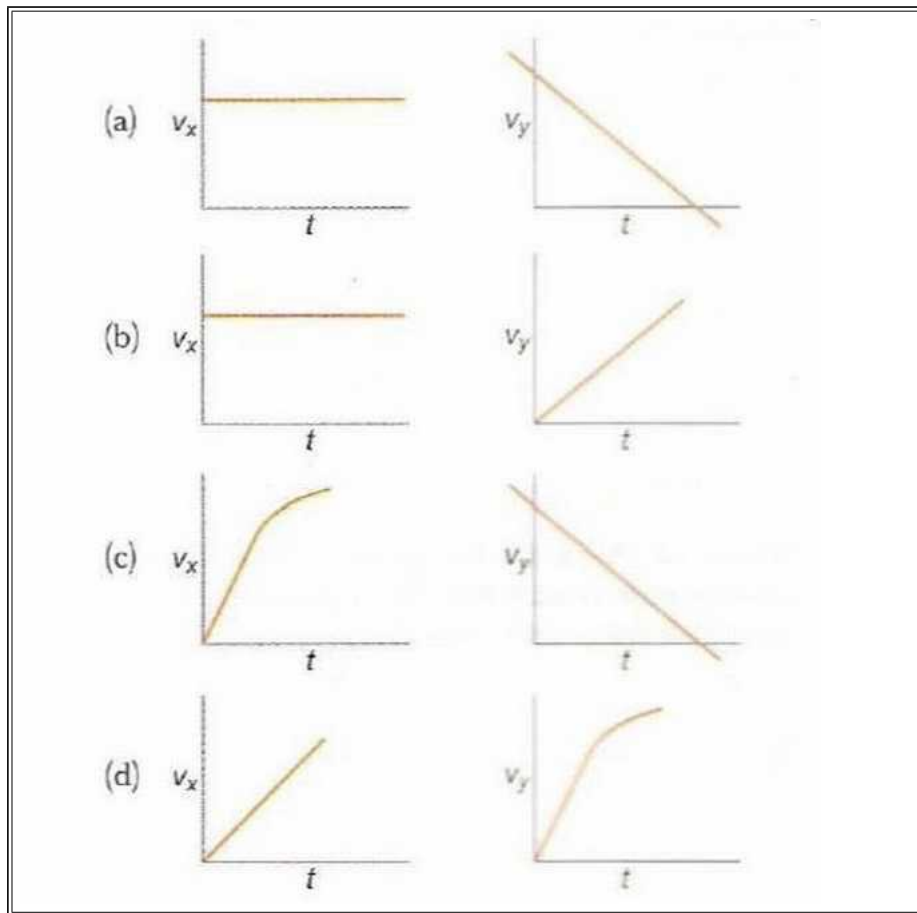
**Example:** A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of  $h$ . Simultaneously another body B is dropped from height  $h$ . It strikes the ground and doesn't rebound. The velocity of A relative to B vs time graph is best represented by (upward direction is positive.)



### 1.2.5 Projectile Motion

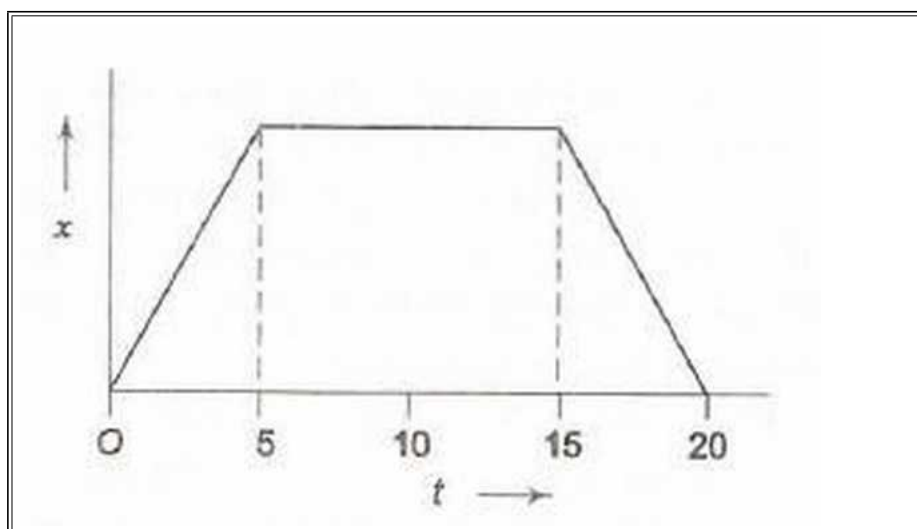
**Example:** A shell is fired from a gun at an angle to the horizontal. Graphs are drawn for its horizontal component of velocity  $v_x$  and its vertical component of velocity  $v_y$ .

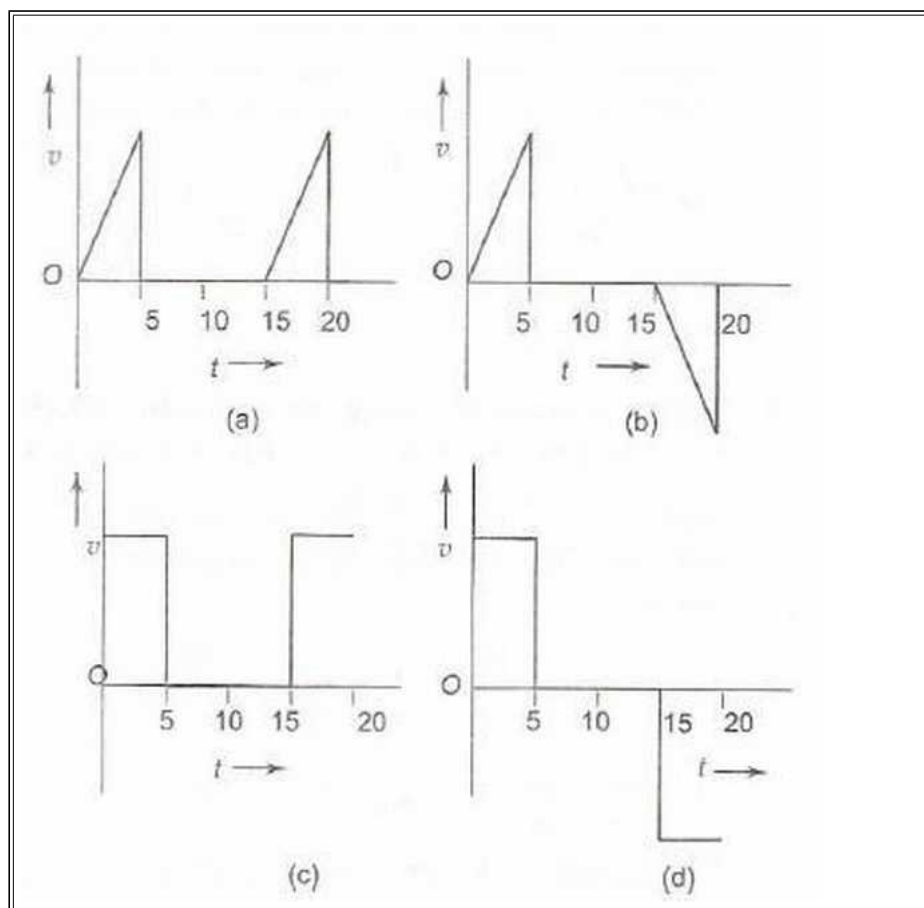




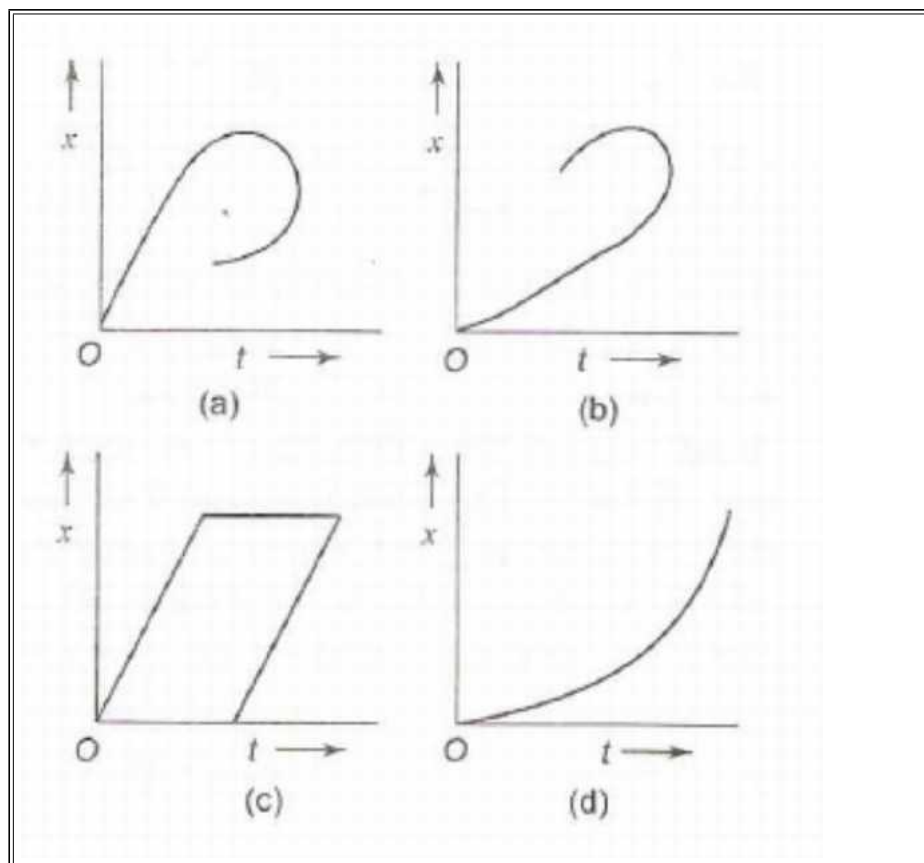
### 1.2.6 Miscellaneous

**Example.** Figure shows the displacement-time ( $x-t$ ) graph of body moving in a straight line. Which one of the graphs shown in Fig. represents the velocity-time ( $v-t$ ) graph of the motion of the body.

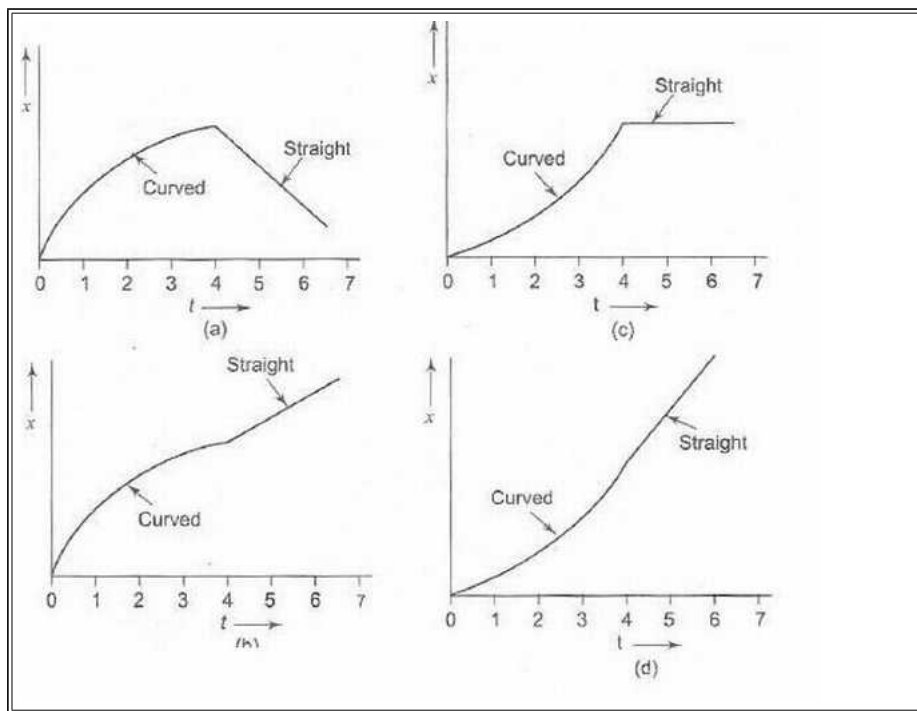




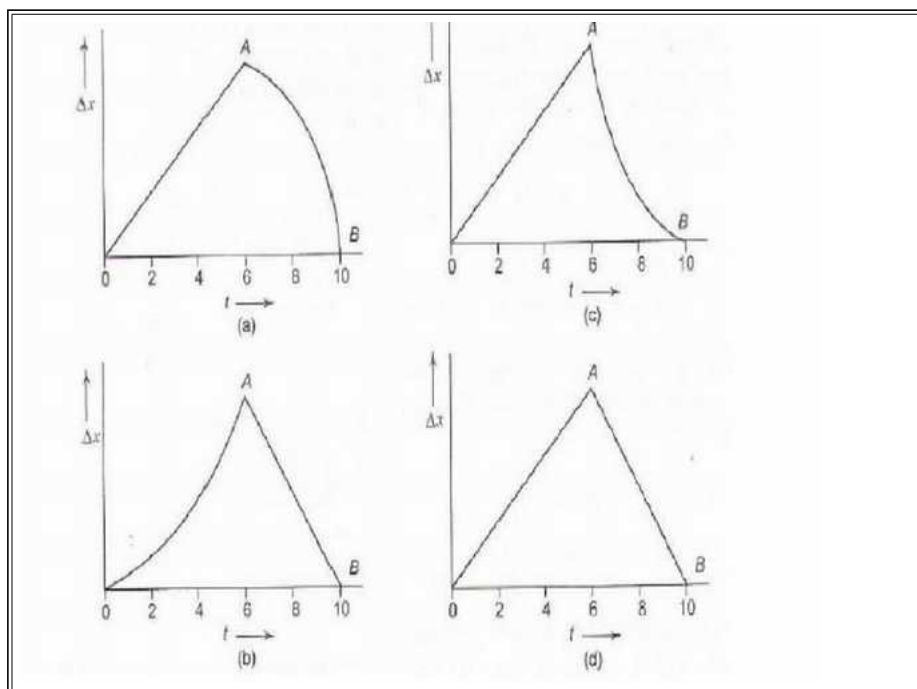
**Example.** Which of the displacement-time ( $x-t$ ) graphs shown in Fig. can possibly represent one dimensional motion of a particle?



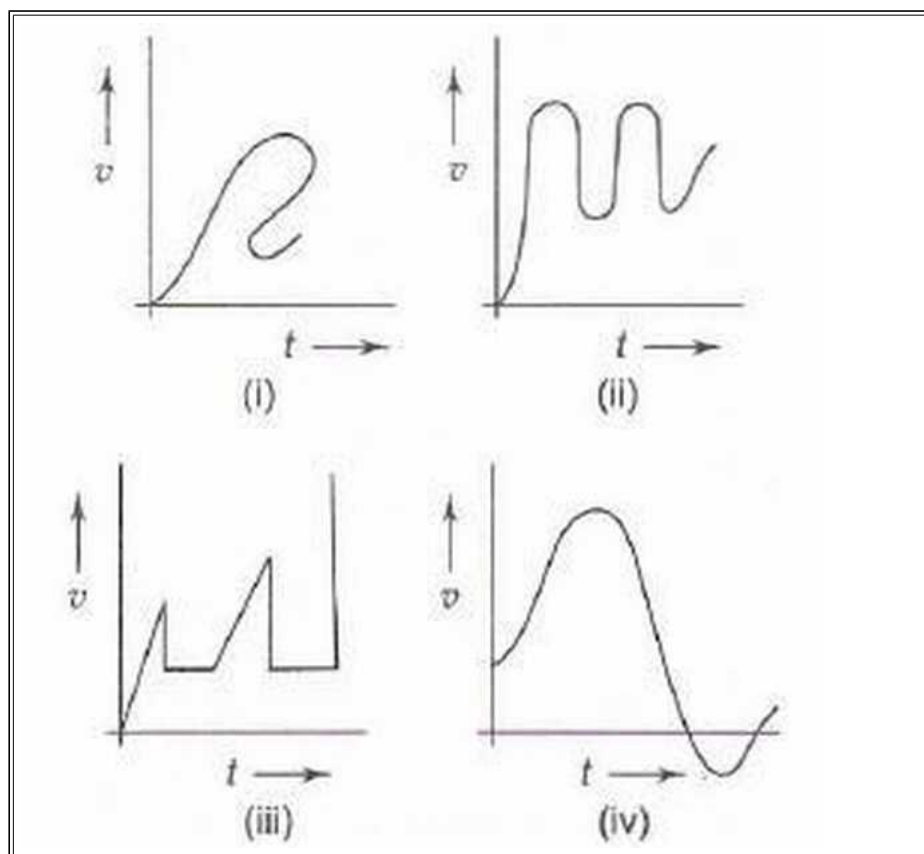
**Example.** A car starts from rest, accelerates uniformly for 4 seconds and then moves with uniform velocity. Which of the ( $x-t$ ) graphs shown in Fig. represents the motion of the car upto  $t = 7$  s?



**Example.** Two stones are thrown up simultaneously with initial speeds of  $u_1$  and  $u_2$ , ( $u_2 > u_1$ ). They hit the ground after 6 s and 10 s respectively. Which graph in Fig. correctly represents the time variation of  $\Delta x = x_2 - x_1$ , the relative position of the second stone with respect to the first upto  $t = 10$  s? Assume that the stones do not rebound after hitting the ground.

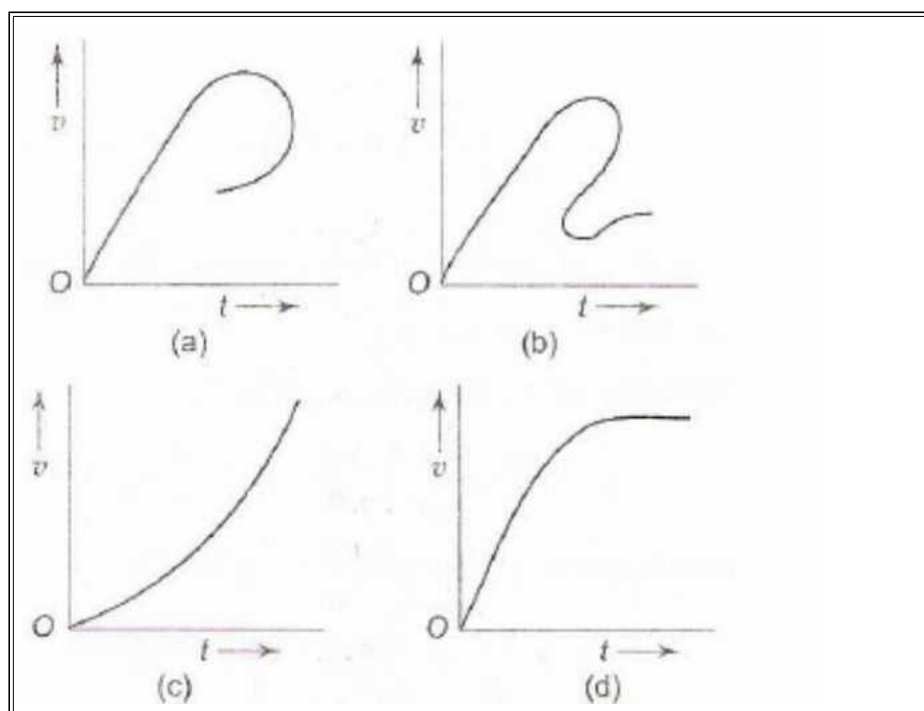


**Example.** Figure shows the velocity-time ( $v - t$ ) graphs for one dimensional motion. But only some of these can be realized in practice. These are



- a) (i), (ii) and (iv) only
- b) (i), (ii) and (iii) only
- c) (ii) and (iv) only
- d) all

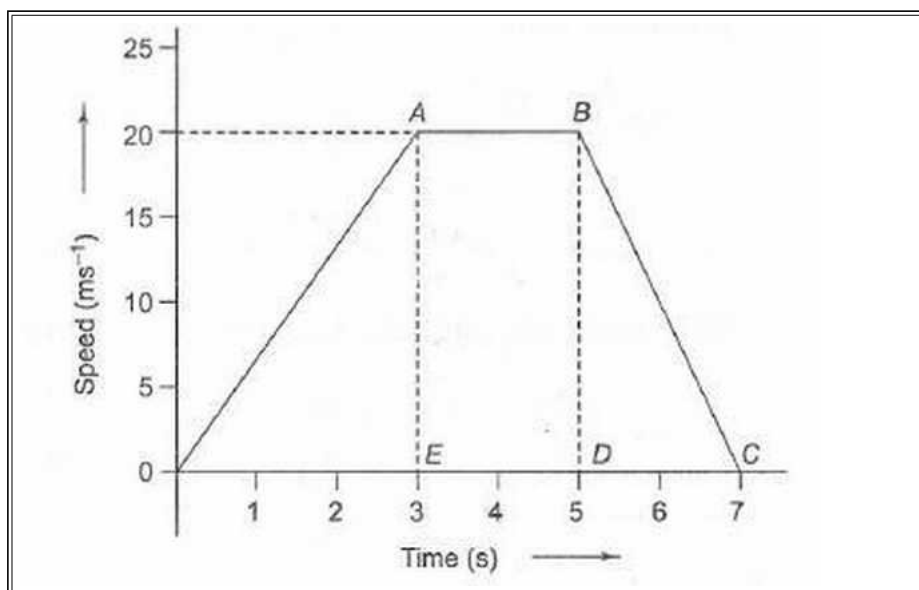
**Example.** Which of the velocity-time ( $v$ - $t$ ) graphs shown in Fig. can possibly represent one-dimensional motion of a particle?



## 1.3 Question Types

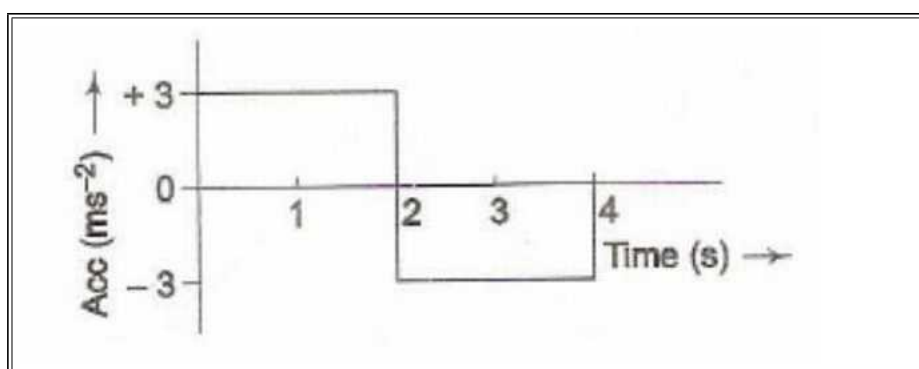
### 1.3.1 Passage Type

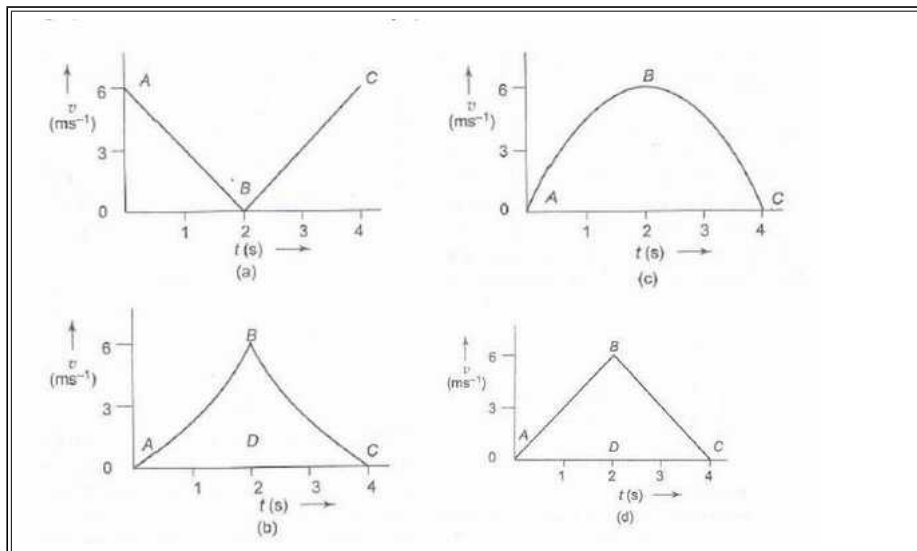
**Example:** The speed-time graph of the motion of a body is shown in Fig.



- The accelerations of the body during the last 2 second is
  - $\frac{20}{3}ms^{-2}$
  - $-\frac{20}{7}ms^{-2}$
  - $-10ms^{-2}$
  - Zero
- The ratio of distance travelled by the body during the last 2 seconds to the total distance travelled by it is
  - 1/9
  - 2/9
  - 3/9
  - 4/9
- The average speed of the car during the whole journey is
  - 10 m/s
  - 20 m/s
  - $90/7$  m/s
  - $40/7$  m/s

**Example:** A body starts from rest at time  $t = 0$  and undergoes an acceleration as shown in Fig. Which of the graphs shown in Fig. represents the velocity-time ( $v-t$ ) graph of the motion of the body from  $t = 0$  s to  $t = 4$  s?





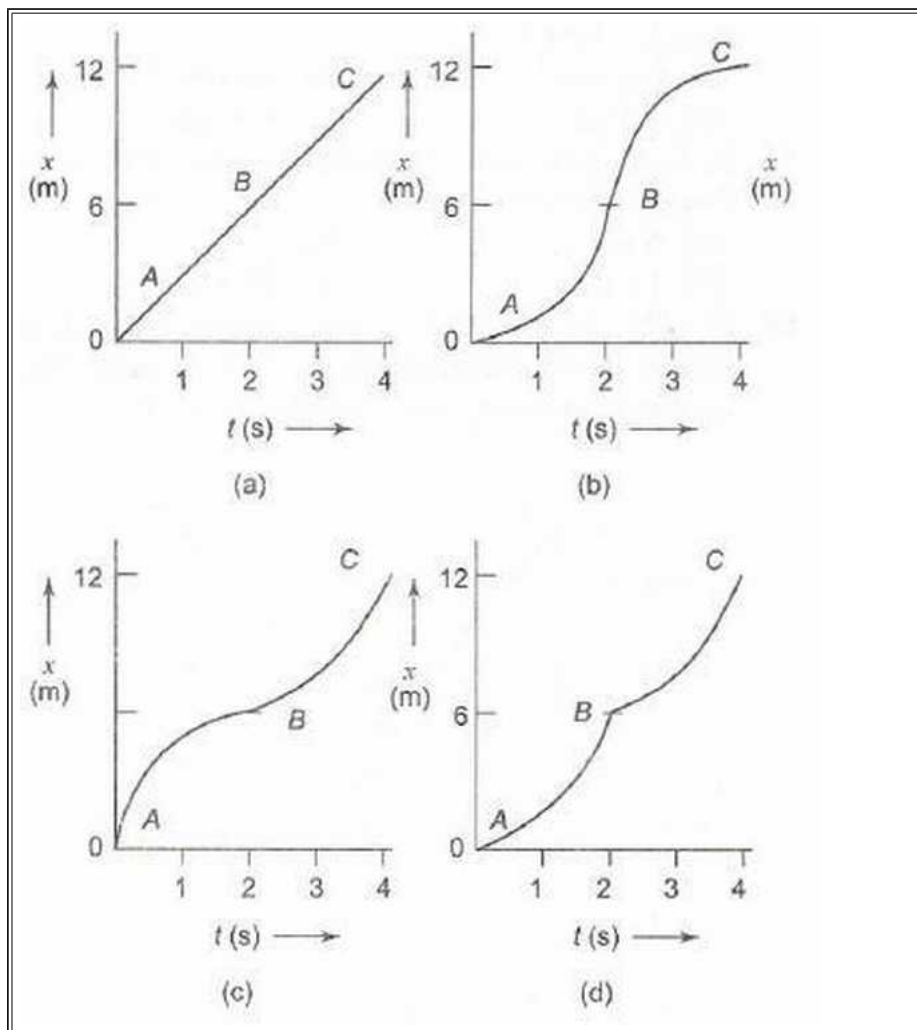
1. In Question above, what is the velocity of the body at time  $t = 2.5$  s?

- a) 2.5 m/s
- b) 3.5 m/s
- c) 4.5 m/s
- d) 5.5 m/s

2. In above question, how much distance does the body cover from  $t = 0$  s to  $t = 4$  s?

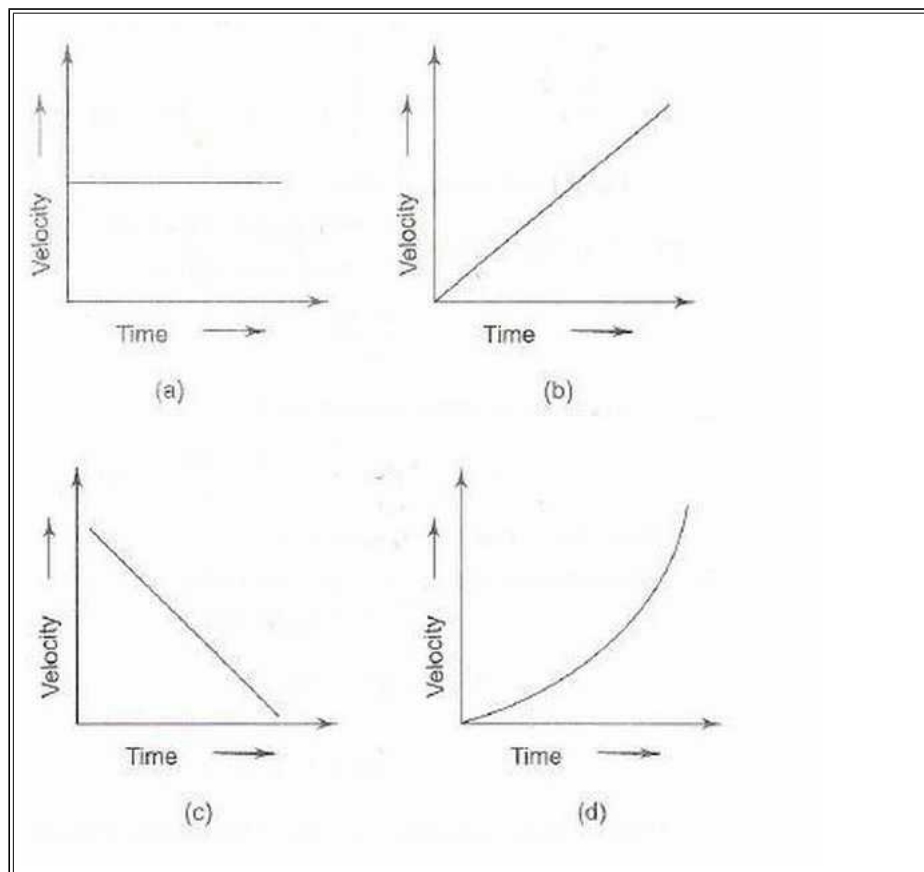
- a) 6 m
- b) 9 m
- c) 12 m
- d) 15 m

3 In above question, which of the graphs shown in Fig. represents the displacement-time ( $x$ - $t$ ) graph of the motion of the body from  $t = 0$  s to  $t = 4$  s?



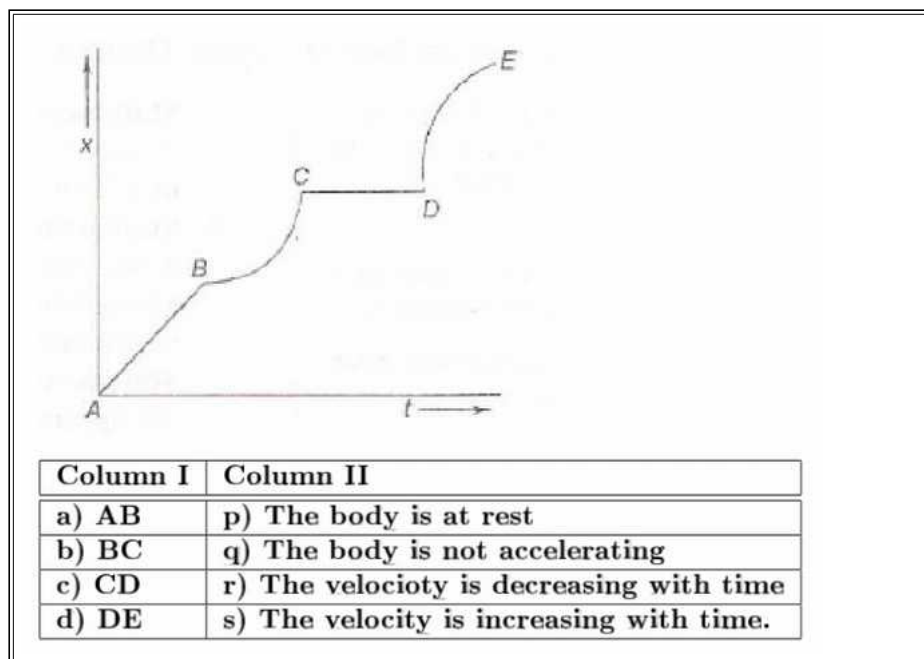
### 1.3.2 Matching

- Match the graphs (a), (b), (c) and (d) shown in Fig. with the types of motions (p), (q), (r) and (s) that they represent



- p) Motion with non-uniform acceleration
- q) Motion of a body covering equal distances in equal intervals of time
- r) Motion having a constant retardation
- s) Uniformly accelerated motion.

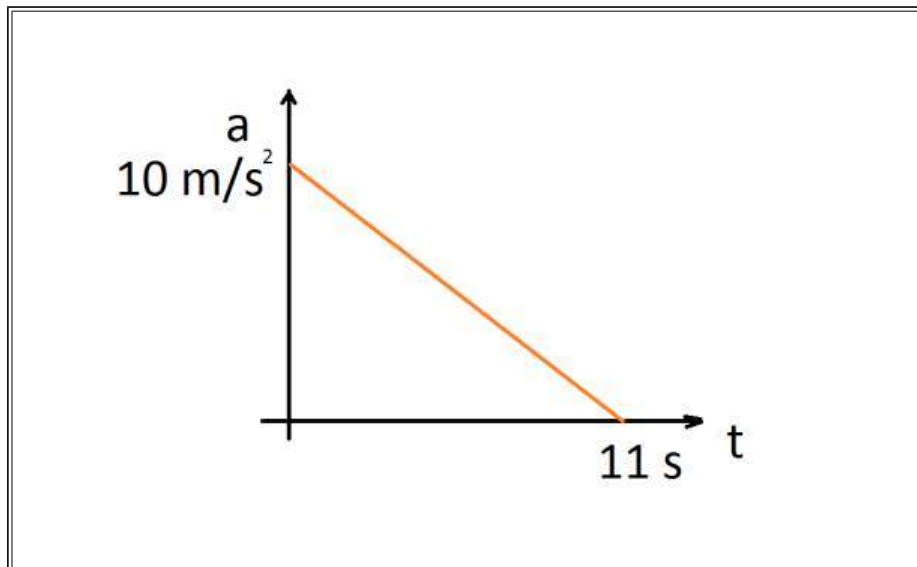
2. Figure shows the displacement-time ( $x-t$ ) graph of the motion of a body.



## 1.4 Previous Year Problems IIT

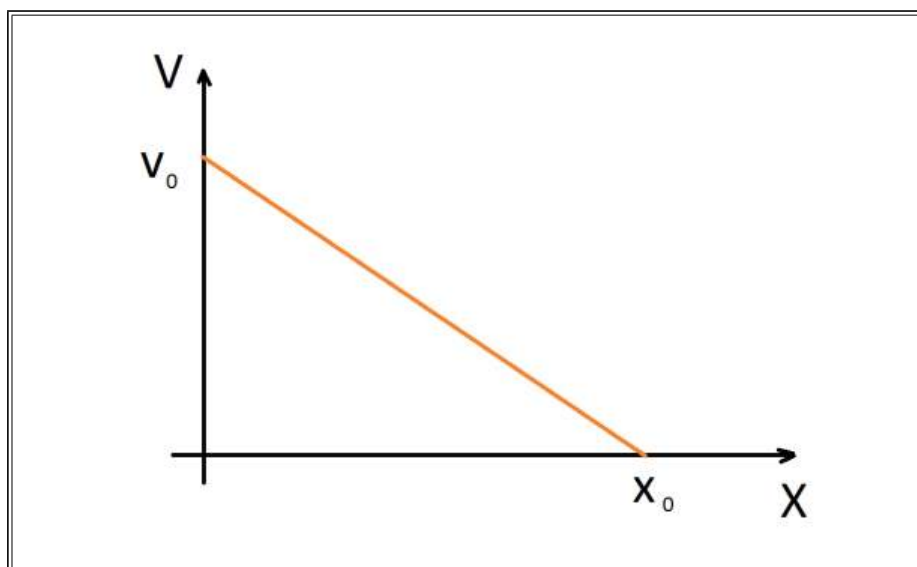
**Q1:** A particle starts from rest. Its acceleration ( $a$ ) versus time ( $t$ ) is as shown in the figure. The maximum speed of the particle will be

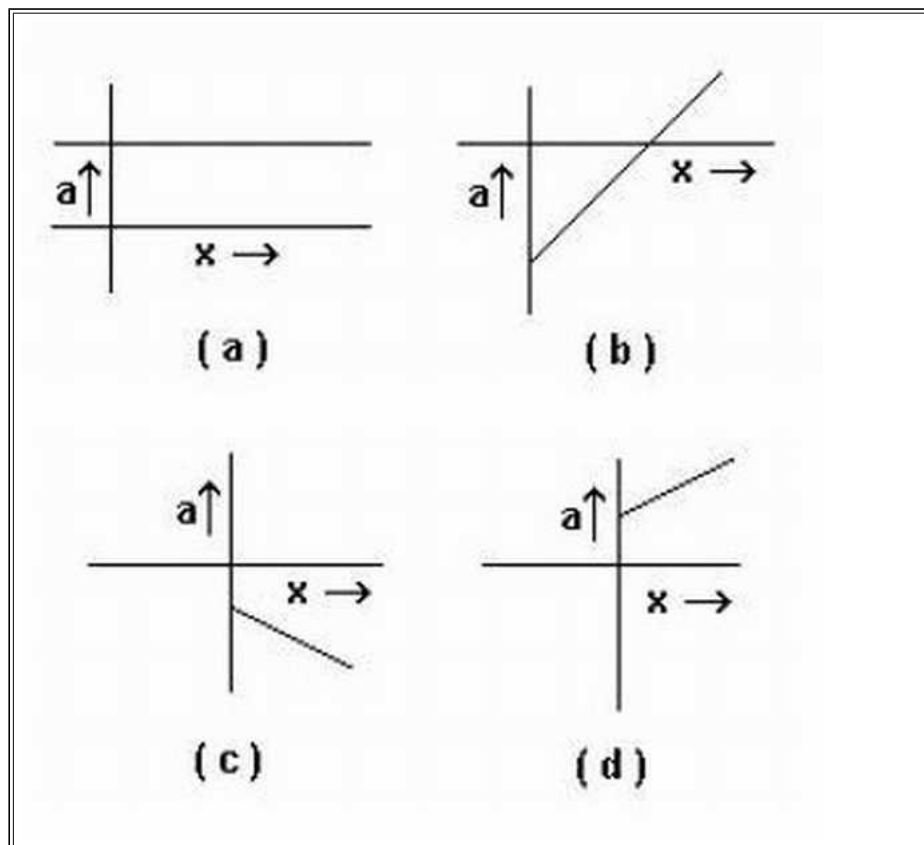




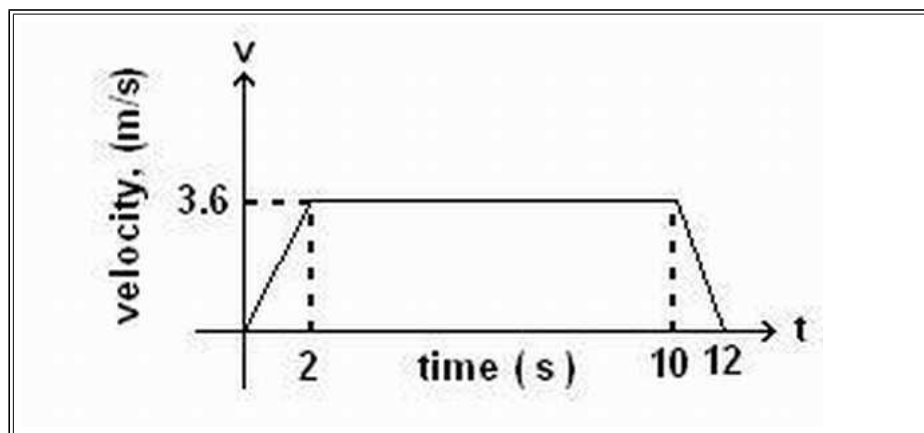
- a )  $110 \text{ m/s}$
- b )  $55 \text{ m/s}$
- c )  $550 \text{ m/s}$
- d )  $660 \text{ m/s}$

**Q2:** If graph of velocity vs. distance is as shown, which of the following graphs correctly represents the variation of acceleration with displacement ?





**Passage** A lift is going up. The variation in the speed of the lift is as given in the graph.



**Q3:** What is the height to which the lift takes the passengers ?

- a ) 3.6 m
- b ) 28.8 m
- c ) 36 m
- d ) cannot be calculated from the above graph

**Q4:** In the above graph, what is the average velocity of the lift?

- a ) 1 m/s
- b ) 2.88 m/s
- c ) 3.24 m/s
- d ) 3 m/s

**Q5:** In the above graph , what is the average acceleration of the lift?

- a)  $1.8m/s^2$
- b)  $-1.8m/s^2$
- c)  $0.3m/s^2$
- d) zero

**Q6:** Four persons K, L, M and N are initially at the corners of a square of side of length  $d$ . If every person starts moving with velocity  $v$  such that K is always headed towards L, L towards M, M towards N and N towards K, then the four persons will meet after

- a)  $d/v$  s
- b)  $d^2/v$  s
- c)  $d/2v$  s
- d)  $d/2v$  s

**Example.** A ball is dropped vertically from a height  $h$  above the ground. It hits the ground and bounces up vertically to a height  $h/2$ . Neglecting subsequent motion and air resistance, its velocity  $v$  varies with the height  $h$  as (see Fig.) (I.I.T. 2000)

